Foundations of Quaternionic Quantum Computing:
Qubits, Spinors, and SU(2) Geometry

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Abstract

This paper assembles a conservative mathematical foundation for the use of quaternions in single-qubit quantum computing. The goal is not to replace standard quantum mechanics, but to organize familiar structures in a way that makes their geometry more explicit. We distinguish carefully between normalized state vectors in $\mathbb{C}^2 \cong \mathbb{R}^4$, the unit sphere S^3 inside that space, pure states modulo global phase, the projective space $\mathbb{CP}^1 \cong S^2$, the single-qubit gate group U(2), and SU(2) representatives obtained after removing global phase. Within that framework, we exhibit an explicit isomorphism between the unit quaternions and SU(2), explain why unit quaternions are therefore a natural model for single-qubit gates modulo global phase, and show how quaternion multiplication exposes the axis–angle and composition structure of rotations. Standard facts are separated from the organizational choices adopted in the RQM Technologies series. The paper is restricted to single-qubit structure and makes no claim about new physics, multi-qubit quaternionic state evolution, or computational advantage.

Keywords: qubit geometry; spinors; SU(2); unit quaternions; Bloch sphere; single-qubit gates

1 Introduction

Single-qubit quantum mechanics is usually presented in matrix language: states are vectors in $\mathbb{C}^2$, observables are Hermitian matrices, and gates are unitary matrices. That formulation is complete and standard. The issue addressed here is not a defect in quantum mechanics itself, but a question of representation. In particular, some geometric features of single-qubit dynamics are more transparent when one passes from raw 2×2 complex matrices to the equivalent language of unit quaternions.

The connection is classical. A normalized two-component spinor is a point on the unit sphere in $\mathbb{C}^2 \cong \mathbb{R}^4$, hence on S^3. The group SU(2) is also diffeomorphic to S^3. Unit quaternions form another model of the same manifold and, more strongly, a Lie-group model of SU(2) [3, 4, 6]. For single-qubit gates this is not an exotic reformulation: it is standard mathematics attached to the well-known double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

The main value of the quaternionic presentation is geometric bookkeeping. In matrix form, one can certainly write

$$U = \cos(\theta/2)I - i\sin(\theta/2)\,(\hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma}),$$

but the composition law remains hidden inside matrix multiplication. In quaternion form, the same gate is written as

$$q = \cos(\theta/2) + \sin(\theta/2)\,(n_x\mathbf{i} + n_y\mathbf{j} + n_z\mathbf{k}),$$

and the product law immediately exhibits dot and cross products. This makes the rotation structure of single-qubit gates unusually explicit.

The scope of this paper is deliberately narrow. It treats only the mathematics needed to explain why quaternions belong naturally in *single-qubit* quantum computing. It does not argue for quaternionic Hilbert spaces as a new physical theory, and it does not claim any hardware or complexity-theoretic advantage. The purpose is foundational exposition: to present standard structures with clear distinctions, and to identify exactly which interpretive or organizational choices belong to RQM Technologies rather than to established mathematics.

2 Mathematical preliminaries

2.1 Qubits as normalized vectors in $\mathbb{C}^2$

A pure single-qubit state is represented by a nonzero vector

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{C}^2,$$

with the physical state unchanged under multiplication by a nonzero complex scalar. After normalization one may assume

$$|\alpha|^2 + |\beta|^2 = 1.$$

The normalized representatives are often called normalized spinors.

Writing

$$\alpha = a + ib, \qquad \beta = c + id, \qquad a, b, c, d \in \mathbb{R},$$

identifies $\mathbb{C}^2$ with $\mathbb{R}^4$.

Proposition 2.1 (Normalized spinors form S^3)**.** *The set*

$$\left\{(\alpha, \beta) \in \mathbb{C}^2 : |\alpha|^2 + |\beta|^2 = 1\right\}$$

is the unit sphere $S^3 \subset \mathbb{R}^4$ under the identification $\mathbb{C}^2 \cong \mathbb{R}^4$.

Proof. Under $\alpha = a + ib$ and $\beta = c + id$, the normalization condition becomes

$$a^2 + b^2 + c^2 + d^2 = 1,$$

which is exactly the defining equation of the unit 3-sphere in $\mathbb{R}^4$. □

Remark 2.2. This S^3 is a space of *normalized representatives*. It is not yet the physical pure-state space, because global phase has not been quotiented out.

2.2 Pure states, global phase, and the Bloch sphere

Two normalized vectors $|\psi\rangle$ and $e^{i\chi}|\psi\rangle$ represent the same pure state. The pure-state space is therefore the projective quotient

$$\mathbb{CP}^1 = (\mathbb{C}^2 \setminus \{0\})/\mathbb{C}^\times,$$

or, after normalization, the quotient of S^3 by the circle action

$$(\alpha, \beta) \mapsto (e^{i\chi}\alpha, e^{i\chi}\beta).$$

Proposition 2.3 (Pure single-qubit states are $\mathbb{CP}^1 \cong S^2$)**.** *The pure-state space of a single qubit is $\mathbb{CP}^1$, and $\mathbb{CP}^1$ is diffeomorphic to S^2. Under this identification the Bloch vector of*

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

is

$$\boldsymbol{r}(\psi) = \left(2\,\mathrm{Re}(\alpha^*\beta),\, 2\,\mathrm{Im}(\alpha^*\beta),\, |\alpha|^2 - |\beta|^2\right) \in S^2.$$

Proof. The quotient description of $\mathbb{CP}^1$ is standard. For normalized $|\psi\rangle$, a direct calculation shows

$$\|\boldsymbol{r}(\psi)\|^2 = 4\,|\alpha^*\beta|^2 + \left(|\alpha|^2 - |\beta|^2\right)^2 = \left(|\alpha|^2 + |\beta|^2\right)^2 = 1.$$

The map $|\psi\rangle \mapsto \boldsymbol{r}(\psi)$ is invariant under $|\psi\rangle \mapsto e^{i\chi}|\psi\rangle$, so it descends to the projective quotient and yields the usual identification of $\mathbb{CP}^1$ with the Bloch sphere S^2 [2, 5]. □

Remark 2.4. The Bloch sphere is therefore a quotient of the state sphere S^3. It is *not* the same object as S^3, and conflating the two obscures the role of global phase.

2.3 Single-qubit gates in $\mathrm{U}(2)$ and representatives in $\mathrm{SU}(2)$

Single-qubit gates are unitary operators on $\mathbb{C}^2$, hence elements of $\mathrm{U}(2)$. Because a global phase on the state vector is physically irrelevant, two gates that differ by an overall phase act identically on pure states.

Proposition 2.5 (Removing global phase from a single-qubit gate)**.** *For every $U \in \mathrm{U}(2)$ there exist $\chi \in \mathbb{R}$ and $V \in \mathrm{SU}(2)$ such that*

$$U = e^{i\chi}V.$$

The representative V is unique up to multiplication by -1.

Proof. Since $\det U \in \mathrm{U}(1)$, choose χ such that $e^{2i\chi} = \det U$, and define $V = e^{-i\chi}U$. Then

$$\det V = e^{-2i\chi}\det U = 1,$$

so $V \in \mathrm{SU}(2)$. If also $U = e^{i\chi'}V'$ with $V' \in \mathrm{SU}(2)$, then

$$V' = e^{i(\chi-\chi')}V.$$

Taking determinants gives $e^{2i(\chi-\chi')} = 1$, hence $e^{i(\chi-\chi')} = \pm 1$ and $V' = \pm V$. □

Thus physically distinct single-qubit gate actions are represented projectively by

$$\mathrm{U}(2)/\mathrm{U}(1) \cong \mathrm{SU}(2)/\{\pm I\} \cong \mathrm{SO}(3),$$

while it is often convenient to work with a chosen representative in $\mathrm{SU}(2)$ before passing to the projective quotient.

2.4 Unit quaternions

Definition 2.6. *The quaternion algebra $\mathbb{H}$ is the real associative algebra with basis $\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ and multiplication rules*

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1.$$

Every quaternion has the form

$$q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}, \qquad a, b, c, d \in \mathbb{R}.$$

Its conjugate is $\overline{q} = a - b\mathbf{i} - c\mathbf{j} - d\mathbf{k}$, and its norm is

$$|q| = \sqrt{q\overline{q}} = \sqrt{a^2 + b^2 + c^2 + d^2}.$$

The unit quaternions are

$$\mathrm{Sp}(1) = \{q \in \mathbb{H} : |q| = 1\}.$$

As a manifold, $\mathrm{Sp}(1)$ is again S^3.

Remark 2.7. At this point two appearances of S^3 coexist:

(i) normalized spinors in $\mathbb{C}^2$, and

(ii) unit quaternions, equivalently $\mathrm{Sp}(1)$.

They are diffeomorphic manifolds, but they play different semantic roles: the first is a space of normalized vectors, while the second is a group of transformations. The distinction matters throughout the paper.

3 The $\mathrm{SU}(2)$–unit-quaternion correspondence

3.1 An explicit map

There are several equivalent conventions for identifying $\mathbb{H}$ with a real subalgebra of 2×2 complex matrices. In this paper we choose the convention

$$\Phi(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = \begin{pmatrix} a - di & -c - bi \\ c - bi & a + di \end{pmatrix}. \tag{1}$$

Equivalently,

$$\Phi(q) = aI - i\big(b\sigma_x + c\sigma_y + d\sigma_z\big),$$

where $\sigma_x, \sigma_y, \sigma_z$ are the Pauli matrices. This convention is convenient because the quaternion basis units align directly with the x-, y-, and z-rotation axes:

$$\Phi(\mathbf{i}) = -i\sigma_x, \qquad \Phi(\mathbf{j}) = -i\sigma_y, \qquad \Phi(\mathbf{k}) = -i\sigma_z.$$

Proposition 3.1 (Unit quaternions are isomorphic to $\mathrm{SU}(2)$)**.** *The restriction of Φ to $\mathrm{Sp}(1)$ is a Lie-group isomorphism*

$$\Phi : \mathrm{Sp}(1) \xrightarrow{\sim} \mathrm{SU}(2).$$

Proof. The assignment is determined by

$$\mathbf{i} \mapsto -i\sigma_x, \qquad \mathbf{j} \mapsto -i\sigma_y, \qquad \mathbf{k} \mapsto -i\sigma_z.$$

These matrices satisfy the same multiplication rules as $\mathbf{i}, \mathbf{j}, \mathbf{k}$:

$$(-i\sigma_x)^2 = (-i\sigma_y)^2 = (-i\sigma_z)^2 = -I,$$

and

$$(-i\sigma_x)(-i\sigma_y) = -i\sigma_z,$$

with cyclic permutations. Hence Φ is an algebra homomorphism and therefore a group homomorphism on invertible elements.

For $q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$, direct calculation gives

$$\det \Phi(q) = a^2 + b^2 + c^2 + d^2 = |q|^2 .$$

If $|q| = 1$, then $\det \Phi(q) = 1$ and $\Phi(q)^\dagger \Phi(q) = I$, so $\Phi(q) \in \mathrm{SU}(2)$. Conversely, any matrix in $\mathrm{SU}(2)$ has the form

$$\begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1,$$

and writing $\alpha = a - di$, $\beta = -c - bi$ recovers a unique unit quaternion $a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$. Smoothness of Φ and Φ^{-1} is immediate from the coordinate formula. □

Corollary 3.2 (Composition corresponds to quaternion multiplication)**.** *If $q_1, q_2 \in \mathrm{Sp}(1)$ represent single-qubit gates through (1), then sequential application of the gates corresponds to quaternion multiplication:*

$$\Phi(q_2 q_1) = \Phi(q_2)\Phi(q_1).$$

Proof. This is the homomorphism property of Φ. □

3.2 Why this is the right language for gates modulo global phase

By the previous section, every physical single-qubit gate class may be represented by an element of $\mathrm{SU}(2)$, unique up to sign, and $\mathrm{SU}(2)$ is modeled exactly by the unit quaternions. It follows that unit quaternions are a natural representation of single-qubit gates *after* global phase has been factored out at the level of $\mathrm{U}(2)$.

Remark 3.3. This statement must be kept precise. Gates live fundamentally in $\mathrm{U}(2)$. The quaternion model does not replace $\mathrm{U}(2)$; rather, it provides a convenient model of the $\mathrm{SU}(2)$ representative chosen after removing overall phase. The remaining identification $q \sim -q$ corresponds to the fact that $\pm V \in \mathrm{SU}(2)$ induce the same rotation on the Bloch sphere.

4 Quaternionic geometry and rotation structure

4.1 Axis–angle form

Every $V \in \mathrm{SU}(2)$ can be written as

$$V = \cos(\theta/2)I - i\sin(\theta/2)(\hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma}), \tag{2}$$

where $\hat{\boldsymbol{n}} = (n_x, n_y, n_z) \in S^2$ and $\theta \in \mathbb{R}$. Under (1), the corresponding quaternion is

$$q = \cos(\theta/2) + \sin(\theta/2)(n_x\mathbf{i} + n_y\mathbf{j} + n_z\mathbf{k}). \tag{3}$$

This is the standard axis–angle form of a unit quaternion [3, 4].

The familiar Pauli rotations are immediate:

$$\begin{aligned} R_x(\theta) &\leftrightarrow \cos(\theta/2) + \sin(\theta/2)\mathbf{i}, \\ R_y(\theta) &\leftrightarrow \cos(\theta/2) + \sin(\theta/2)\mathbf{j}, \\ R_z(\theta) &\leftrightarrow \cos(\theta/2) + \sin(\theta/2)\mathbf{k}. \end{aligned}$$

In particular, π-rotations correspond, up to global phase, to the basis units $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

4.2 Composition as scalar–vector algebra

Write a quaternion as $q = a + \mathbf{u}$, where $a \in \mathbb{R}$ and $\mathbf{u} = b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ is its pure part, identified with a vector in $\mathbb{R}^3$. Then quaternion multiplication becomes

$$(a_1 + \mathbf{u}_1)(a_2 + \mathbf{u}_2) = \big(a_1a_2 - \mathbf{u}_1 \cdot \mathbf{u}_2\big) \ + \ \big(a_1\mathbf{u}_2 + a_2\mathbf{u}_1 + \mathbf{u}_1 \times \mathbf{u}_2\big). \tag{4}$$

Equation (4) makes three features transparent:

(a) the scalar part records the dot product of the pure parts;

(b) the pure part contains the cross product responsible for noncommutativity; and

(c) sequential gate composition is visibly a rotation-composition law, not merely an abstract matrix product.

This is one sense in which quaternionic notation clarifies geometry that matrix notation can hide. The matrix formulation already contains the same information, but it is encoded less directly.

4.3 The double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$

The action of a unit quaternion on a pure quaternion $v = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ by conjugation,

$$v \longmapsto qvq^{-1},$$

preserves the pure-quaternion subspace and its Euclidean norm. Identifying pure quaternions with $\mathbb{R}^3$ therefore gives a homomorphism

$$\rho : \mathrm{Sp}(1) \to \mathrm{SO}(3).$$

Proposition 4.1 (Unit quaternions double-cover SO(3))**.** *The map ρ is surjective and has kernel $\{\pm 1\}$. Consequently,*

$$\mathrm{Sp}(1)/\{\pm 1\} \cong \mathrm{SO}(3).$$

Via Φ, the same statement is the familiar double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

Proof. The conjugation action preserves multiplication and the norm on pure quaternions, hence lands in SO(3). Standard rotation theory shows that every element of SO(3) admits an axis–angle lift of the form (3), establishing surjectivity. If $qvq^{-1} = v$ for every pure quaternion v, then q commutes with all pure quaternions, which forces $q \in \mathbb{R}$. Since q is unitary, $q = \pm 1$. □

Remark 4.2. The sign ambiguity here should not be confused with global phase for state vectors, although the two are related. A 2π rotation in SO(3) lifts to a path in SU(2) from I to $-I$, so spinors return to the same physical pure state only projectively. This is standard spin-1/2 geometry [6].

5 What is standard mathematics, and what is RQM framing?

5.1 Standard facts used in this paper

The following statements are standard and not specific to RQM Technologies:

(1) normalized vectors in $\mathbb{C}^2$ form S^3;

(2) pure single-qubit states are projective classes, giving $\mathbb{CP}^1 \cong S^2$;

(3) single-qubit gates live in U(2);

(4) after removal of overall phase, gates admit representatives in SU(2);

(5) the unit quaternions form a Lie group isomorphic to SU(2);

(6) SU(2) is a double cover of SO(3); and

(7) axis–angle rotations can be written equivalently in Pauli-matrix form or quaternion form.

These are standard results from quantum mechanics, Lie theory, and quaternionic rotation theory [3–6].

5.2 RQM Technologies framing choices

The project-specific content of this paper is modest and organizational:

(a) **Primary expository emphasis.** The paper chooses the quaternion model as the preferred language for describing *single-qubit gate geometry*.

(b) **Convention choice.** The explicit map Φ in (1) is selected so that $\mathbf{i}, \mathbf{j}, \mathbf{k}$ align with the x, y, z Pauli axes. This is one standard convention among several equivalent ones.

(c) **Series organization.** The paper serves as a terminology and notation baseline for later papers in the series.

These are framing choices, not new physical claims. They do not alter the content of standard single-qubit quantum mechanics.

6 Scope, limitations, and non-claims

This paper is intentionally limited. In particular, it makes the following non-claims explicit:

(1) It does *not* claim that standard complex-matrix methods are inadequate or invalid.

(2) It does *not* claim a new physical theory or a replacement of ordinary quantum mechanics.

(3) It does *not* claim a hardware implementation advantage.

(4) It does *not* claim a complexity-theoretic or algorithmic advantage.

(5) It does *not* develop a multi-qubit quaternionic state formalism.

(6) It does *not* treat quaternionic Hilbert-space quantum mechanics in the sense studied by Adler [1]; that subject is distinct from the present gate-level representation question.

The paper should therefore be read as a clarifying foundation for single-qubit language only.

7 Conclusion

For a single qubit, the relevant geometry links together several standard structures:

- normalized spinors in $\mathbb{C}^2$, forming S^3;
- pure states modulo global phase, forming $\mathbb{CP}^1 \cong S^2$;
- single-qubit gates in U(2), with SU(2) representatives after removing global phase; and
- unit quaternions as an explicit Lie-group model of SU(2).

The quaternion model is therefore natural not because it changes the theory, but because it presents the existing rotation structure of single-qubit gates in a compact geometric form. In that sense, quaternions belong naturally in single-qubit quantum computing as a representation language for SU(2) and its relation to the Bloch sphere.

A Notation used repeatedly

Symbol	Meaning	Role
$\lvert\psi\rangle$	normalized state vector in $\mathbb{C}^2$	state representative
S^3	unit sphere in $\mathbb{C}^2 \cong \mathbb{R}^4$	normalized spinors
$\mathbb{CP}^1$	projective pure-state space	pure states modulo phase
S^2	Bloch sphere	pure states / Bloch vectors
U	element of U(2)	single-qubit gate
V	representative in SU(2)	gate modulo global phase
q	unit quaternion	model of $V \in \mathrm{SU}(2)$
Φ	quaternion–matrix identification	convention chosen here

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